

SQL Exercise 1

① SELECTING & SORTING

① SELECT *

FROM employees;

Output:

id	First_name	Last_name	department	Salary	hire_date	city
1	Alice	Green	IT	70000	2020-01-01	Johannesburg
2	Brian	Lee	HR	45000	2019-03-22	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-18	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-09-30	Johannesburg

② SELECT DISTINCT department

FROM employees;

output:

department
IT
HR
Finance
Marketing

③ SELECT first_name, last_name

FROM employees

ORDER BY Salary DESC;

Output:

First_name	Last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

④ SELECT id, first_name, Last_name, Salary
FROM employees
ORDER BY Salary DESC
LIMIT 3;

Output:

id	Firstname	last_name	Salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

② Filtering with WHERE

⑤ SELECT id, first_name, last_name, department
FROM employees
WHERE department = 'IT';

Output:

id	Firstname	last_name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

⑥ SELECT id, first_name, last_name, department, Salary
FROM employees
WHERE department = 'Finance' AND Salary > 60000

Output:

id	First-name	last_name	department	Salary
3	Cathy	Zulu	Finance	65000

⑦ SELECT id, First_name, Last_name, department
 FROM employees
 WHERE department = 'HR' OR department = 'Marketing';

SELECT id, Firstname, Last_name, department
 FROM employees
 WHERE department IN ('HR', 'Marketing');

Output:

id	First_name	Last_name	department
2	Brian	Lee	HR
4	David	Mokoena	Marketing

⑧ SELECT id, First_name, Last_name, department
 FROM employees
 WHERE department != 'IT';

Output:

id	Firstname	Last_name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokoena	Marketing

⑨ SELECT id, first_name, last_name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

Output:

id	first_name	last_name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

⑩ SELECT id, first_name, last_name, department, salary, city
FROM employees
WHERE department = 'IT' AND Salary > 5000 AND city = 'Johannesburg';

id	first_name	last_name	department	Salary	City
1	Alice	Green	IT	70000	Johannesburg
5	Eva	Naidoo	IT	72000	Johannesburg